2 NETWORKING

OSI Model + Transport Layer

D2 — Networking

CYBERLABS

OSI LAYER PDUs (memorise this table)

- Layer 4 Transport → Segments (TCP) / Datagrams (UDP)
- Layer 3 Network → Packets
- Layer 2 Data Link → Frames
- Layer 1 Physical → Bits

EXAM SHORTCUT — L4=Segments, L3=Packets, L2=Frames, L1=Bits

TCP CONTROL FLAGS

- SYN — initiate connection (3-way handshake step 1)
- ACK — acknowledge received data
- RST — hard abort — immediate connection tear-down, no handshake
- FIN — graceful close — both sides agree to end the session

EXAM SHORTCUT — RST = hard reset (error/unexpected packet). FIN = polite goodbye

TLS 1.3 HANDSHAKE + SSH

- TLS: asymmetric crypto (RSA/ECDH) used only to negotiate a session key
- Once session key is agreed → symmetric cipher (AES) handles all data
- Why: asymmetric is slow; symmetric is fast for bulk transfer
- SSH default: TCP port 22 (encrypted, reliable, remote shell)

EXAM SHORTCUT — TLS = asymmetric to share the key, symmetric to use it

OSI Model — 7 Layers

D2 — Networking

CYBERLABS

OSI Model — 7 Layers

SEND ↑

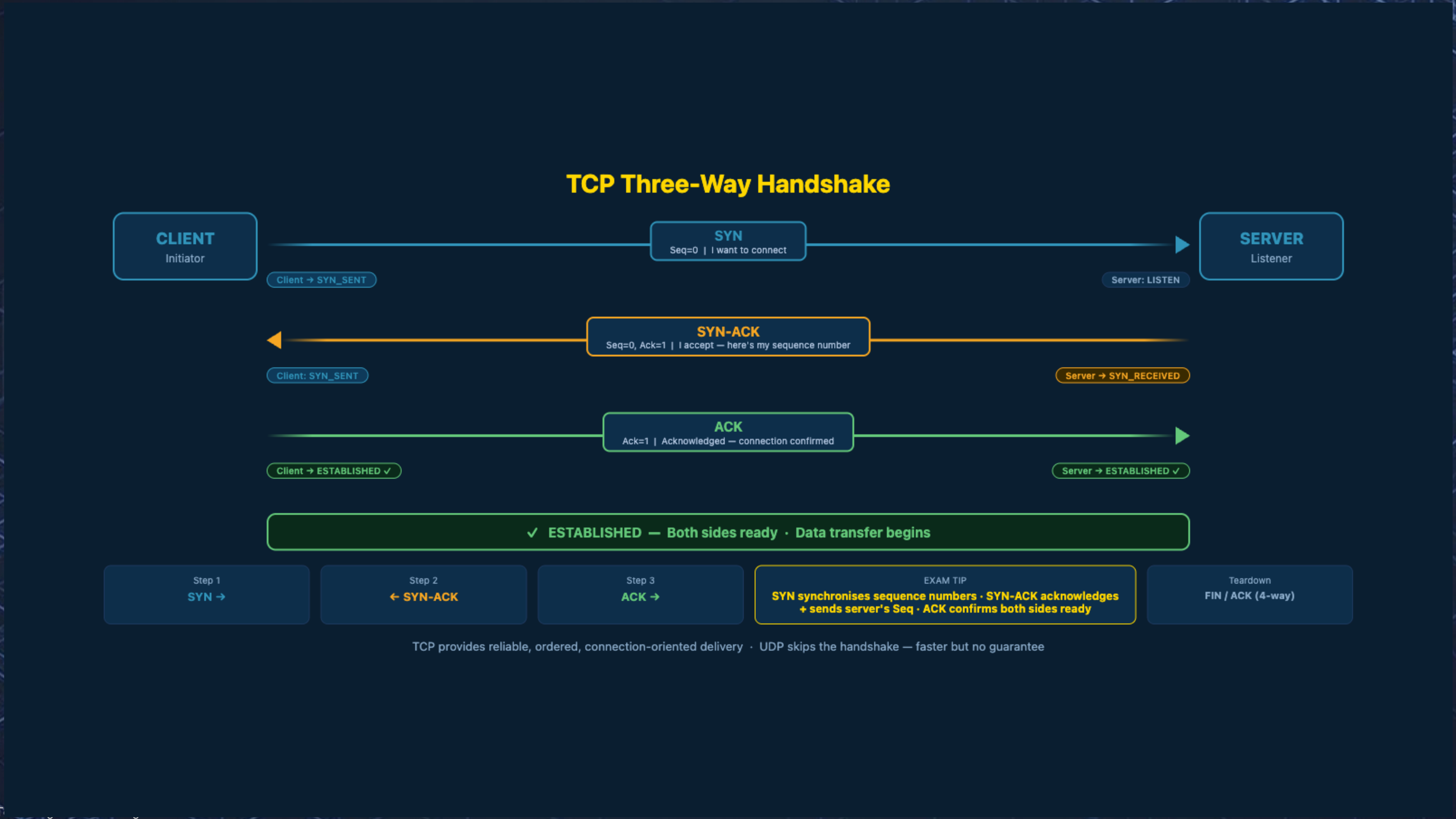
#	LAYER	PROTOCOLS / EXAMPLES	MNEMONIC	PDU NAME
7	Application	HTTP, HTTPS, FTP, SMTP, DNS, SSH, Telnet	*All*	Data
6	Presentation	SSL/TLS, JPEG, MPEG, ASCII — encryption & encoding	*People*	Data
5	Session	NetBIOS, RPC, PPTP — session open/close/sync	*Seem*	Data
4	Transport	TCP (reliable, ordered) · UDP (fast, connectionless) · Ports	*To*	Segment / Datagram
3	Network	IP (v4/v6), ICMP, OSPF, BGP — logical addressing & routing	*Need*	Packet
2	Data Link	Ethernet, Wi-Fi (802.11), MAC addresses, ARP, VLANs, switches	*Data*	Frame
1	Physical	Cables, hubs, repeaters, NIC, 802.11 radio — raw bits on wire	*Processing*	Bits

↓ RECEIVE

Mnemonic (top→bottom): **A**ll **P**eople **S**eem **T**o **N**eed **D**ata **P**rocessing · Encapsulation adds a header at each layer going down; decapsulation strips headers going up

TCP Three-Way Handshake

D2 — Networking



Practice Question 2.1

D2 — Networking

QUESTION 2.1

A packet capture shows a TCP segment with only the RST flag set, sent from a server to a client immediately after the client attempted to connect to a closed port. What does this RST flag tell the client?

A. The server is requesting a graceful session teardown using sequence acknowledgment counters.

B. The server is immediately aborting the connection because the port is closed or the packet was unexpected.

C. The server is initiating a new three-way handshake on a different port.

D. The server is pushing buffered application data to the client without delay.

Answer — Question 2.1

D2 — Networking

ANSWER B

B. The server immediately aborts the connection — the port is closed or the packet was unexpected.

Why this is correct

RST (Reset) is a hard abort signal. Unlike FIN (which is a graceful close), RST immediately terminates the connection with no further exchange. It is sent when a packet arrives at a closed port, or when the receiver determines the session is invalid.

Why the other options are wrong

- A — That describes a FIN flag, not RST. FIN initiates graceful teardown with acknowledgment.
- C — SYN is used to initiate new connections, not RST.
- D — PSH (Push) forces buffered data to the application layer; RST has no relation to data delivery.

IP Addressing — TTL + Subnetting

D2 — Networking

CYBERLABS

TTL (Time to Live)

- A counter in every IPv4 packet header, set by the sender (default 64 or 128)
- Every router that forwards the packet decrements TTL by 1
- TTL reaches 0 → router discards packet + sends ICMP "Time Exceeded" to source
- Purpose: prevents packets from looping forever in misconfigured networks

EXAM SHORTCUT — TTL = loop prevention. Each hop -1. Hits 0 = dropped + ICMP error back

/28 SUBNETTING (exam favourite)

- /28 mask = 255.255.255.240 → block size = 256 - 240 = 16 addresses
- Subnets: .0-.15 | .16-.31 | .32-.47 | .48-.63 …
- First subnet 192.168.10.0/28: Network=.0 | Hosts=.1-.14 | Broadcast=.15
- Usable hosts per /28 subnet = 16 - 2 = 14

EXAM SHORTCUT — /28 block=16. Broadcast = last address (.15, .31, .47...). Usable = 14

Practice Question 2.2

D2 — Networking

QUESTION 2.2

A host is assigned the IP address 172.16.5.20 with a /28 subnet mask. What is the broadcast address of this subnet, and how many usable host addresses does it contain?

A. Broadcast: 172.16.5.31 | 14 usable hosts

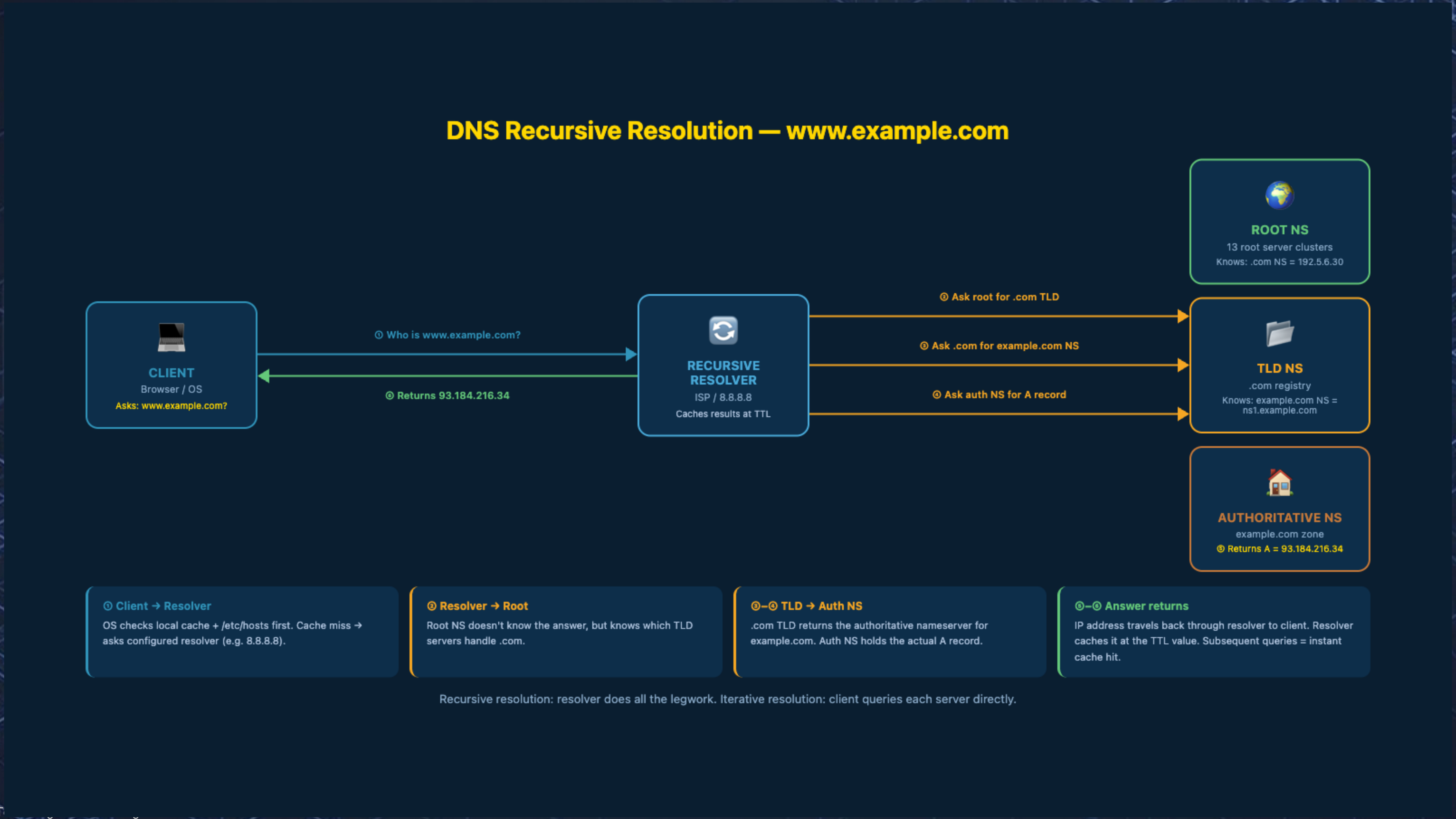
B. Broadcast: 172.16.5.30 | 16 usable hosts

C. Broadcast: 172.16.5.255 | 254 usable hosts

D. Broadcast: 172.16.5.15 | 14 usable hosts

DNS Recursive Resolution

D2 — Networking



Answer — Question 2.2

D2 — Networking

ANSWER A

A. Broadcast: 172.16.5.31 | 14 usable hosts

Why this is correct

/28 block size = 16. The host .20 falls in the block .16-.31. Network address = .16, broadcast = .31, usable = .17 through .30 = 14 hosts.

Why the other options are wrong

- B — .30 is the last usable host, not the broadcast. Broadcast is always the last address in the block.
- C — .255 is the broadcast of a /24 (/28 is much smaller, only 16 addresses per block).
- D — .15 is the broadcast of the first /28 block (.0-.15); this host is in the second block (.16-.31).

Routing + Switching

D2 — Networking

CYBERLABS

802.1Q VLAN TRUNKING

- Standard that allows multiple VLANs to share a single physical trunk link
- Inserts a 4-byte VLAN tag between Source MAC and EtherType in the frame
- 12-bit VLAN ID field supports up to 4094 VLANs (0 and 4095 reserved)

EXAM SHORTCUT — 802.1Q = 4-byte tag inserted after Src MAC. Enables VLAN trunking

STP — SPANNING TREE PROTOCOL

- Layer 2 frames have NO TTL — a loop would create an infinite broadcast storm
- STP detects redundant paths and logically blocks all but one active path
- Blocked ports stay on standby — activated if the active path fails

EXAM SHORTCUT — STP = blocks redundant L2 paths to prevent broadcast storms

Routing + Switching

D2 — Networking

PORT SECURITY — SHUTDOWN MODE

- Unauthorised MAC hits port → interface goes err-disabled immediately
- Violation counter increments; SNMP trap + syslog alert generated
- Recovery: admin must manually re-enable with "no shutdown"

EXAM SHORTCUT — Shutdown mode = err-disabled + SNMP trap. Port stays down until manually fixed

PAT / NAT OVERLOAD

- Maps many private IPs to a single public IP address
- Tracks each session by assigning a unique ephemeral source port number
- State table: [private IP : private port] ↔ [public IP : unique port]

EXAM SHORTCUT — PAT = one public IP, hundreds of sessions separated by port numbers

Practice Question 2.3

D2 — Networking

QUESTION 2.3

A switch running port-security detects a MAC address on a port that is not in the allowed list. The violation mode is configured as "Shutdown". Which sequence of events occurs?

A. The switch silently drops the frame and logs a debug message only.

B. The port is immediately placed into err-disabled state, the violation counter increments, and an SNMP trap is sent.

C. The unauthorised frame is forwarded to a quarantine VLAN for inspection.

D. The switch performs a full configuration reset to remove the port-security policy.

Answer — Question 2.3

D2 — Networking

ANSWER B

B. The port is immediately placed into err-disabled state, the violation counter increments, and an SNMP trap is sent.

Why this is correct

Shutdown is the default and most aggressive port-security violation mode. The interface is disabled (err-disabled state), the link goes down, the violation counter increases, and administrators are alerted via syslog and SNMP. The port must be manually re-enabled with the "no shutdown" command.

Why the other options are wrong

- A — That describes "Protect" mode, which silently drops frames without alerts or disabling the port.
- C — Port-security does not automatically redirect traffic to VLANs; that is a VLAN ACL feature.
- D — Port-security never resets the switch configuration; only the specific interface is affected.

Physical Infrastructure — STP Cable + PoE

D2 — Networking

CYBERLABS

STP CABLING (Shielded Twisted Pair)

- Has a metallic foil or braid shield wrapped around the twisted wire pairs
- Shield absorbs ambient EMI (electromagnetic interference) from the environment
- MUST be properly grounded — the shield bleeds off collected noise to earth
- Without grounding: shield acts as an antenna, amplifying noise into the signal

EXAM SHORTCUT — STP ungrounded = shield becomes antenna. Ground it to drain EMI safely

PoE STANDARDS (Power over Ethernet)

- 802.3af = up to 15.4 W — standard PoE (basic VoIP phones, access points)
- 802.3at = up to 30 W — PoE+ (PTZ cameras, higher-draw devices)
- 802.3bt = up to 90 W — PoE++ (high-performance APs, small switches)
- Delivers both data and DC power over the same twisted pair cable

EXAM SHORTCUT — PoE+ (802.3at) = 30 W. Enough for PTZ cameras and power-hungry devices

Practice Question 2.4

D2 — Networking

QUESTION 2.4

A technician installs STP cabling between two switches in a manufacturing facility that operates large electric motors nearby. After installation, the link experiences intermittent signal errors that were not present before. The cable length and connector crimps are verified correct. What is the most likely root cause?

A. STP cable cannot operate at gigabit speeds in high-EMI environments.

B. The metallic shield was not grounded, causing it to collect and inject EMI into the signal pairs rather than drain it.

C. The cable exceeds the 90-metre maximum length limit for copper Ethernet.

D. STP cables require a PoE-capable switch to activate the shielding layer.

Answer — Question 2.4

D2 — Networking

ANSWER B

B. The metallic shield was not grounded — it collects EMI and injects it into the signal pairs.

Why this is correct

STP cabling uses its shield to absorb EMI from the environment. The shield must be connected to earth ground at the connector junction so accumulated noise drains away. Without that ground path, the shield stores the collected energy and couples it directly into the signal wires — making noise worse, not better.

Why the other options are wrong

- A — STP cable fully supports gigabit speeds; the shielding does not limit throughput.
- C — A length issue would cause consistent errors across all conditions, not intermittent ones tied to nearby motors.
- D — PoE is power delivery over data pairs and has no relationship to shield grounding or EMI protection.

DNS + Network Recon Tools

D2 — Networking

CYBERLABS

DNS RECORD TYPES

- A → hostname maps to an IPv4 address (e.g. www → 1.2.3.4)
- AAAA → hostname maps to an IPv6 address
- CNAME → alias maps to another hostname (canonical name)
- MX → specifies the mail server for a domain
- TXT → free-text (used for SPF, DMARC, domain verification)

EXAM SHORTCUT — A = IPv4 address. CNAME = alias to another name. MX = mail

DNS CACHE POISONING

- Attacker sends forged DNS response before the legitimate server replies
- Forged packet must match the resolver's 16-bit query transaction ID
- Successful poison: resolver caches malicious IP for the domain
- All users querying that resolver get redirected to the attacker's IP

EXAM SHORTCUT — Poisoning = race to inject fake response before real server replies

WIRESHARK + NMAP

- Wireshark ip.addr == X.X.X.X → any traffic involving that IP (src OR dst)
- Wireshark ip.src == X && ip.dst == X → wrong: requires BOTH fields = same IP
- Nmap -sS → SYN scan: sends SYN, reads response, never completes handshake (stealthy)
- Nmap -sV → version scan: probes open ports to identify service and version

EXAM SHORTCUT — ip.addr matches either direction. -sS = stealthy SYN. -sV = version detect

Practice Question 2.5

D2 — Networking

QUESTION 2.5

A SOC analyst is threat-hunting and wants to capture all Wireshark traffic involving a suspicious host at 10.10.10.50 — whether it is the source or the destination. Which display filter is correct?

A. ip.src == 10.10.10.50 && ip.dst == 10.10.10.50

B. ip.addr == 10.10.10.50

C. tcp.host == 10.10.10.50

D. ether.addr == 10.10.10.50

Answer — Question 2.5

D2 — Networking

CYBERLABS

D2 — Networking

ANSWER B

B. ip.addr == 10.10.10.50

Why this is correct

The ip.addr field in Wireshark evaluates both ip.src and ip.dst. It returns any packet where the IP address appears in either direction — exactly what is needed to capture all traffic involving a single host.

Why the other options are wrong

- A — Using && between ip.src and ip.dst requires BOTH source AND destination to equal the same IP, which is impossible in normal traffic. This filter would return zero results.
- C — tcp.host is not a valid Wireshark display filter field.
- D — ether.addr filters on MAC addresses (Layer 2), not IP addresses. It would fail to match routed traffic where the MAC changes at each hop.

Route Summarization + Aggregation

D2 — Networking

WHAT IS ROUTE SUMMARIZATION?

- Combines multiple contiguous subnet routes into a SINGLE routing advertisement
- Router advertises one summary route instead of many specific ones
- Example: 10.2.0.0/24, 10.2.1.0/24, 10.2.2.0/24, 10.2.3.0/24
- → Summarized as: 10.2.0.0/22 (covers all four /24 networks)
- The summary prefix covers the same address space but as one entry

EXAM SHORTCUT — Summarization = fewer routing entries = smaller table = faster lookups

WHY SUMMARIZE?

- Reduces routing table size → less router RAM consumed
- Faster routing lookups — fewer entries to search
- Less routing protocol traffic — one update instead of four
- Hides topology instability: one /24 flapping does not ripple upstream
- Does NOT compress payloads, does NOT use UDP 53, does NOT change IP assignment

EXAM SHORTCUT — Benefit = smaller routing table + faster processing. Nothing to do with DNS or compression.

HOW TO FIND THE SUMMARY ADDRESS

- Align on the bit boundary: find the common bits across all subnets
- 10.2.0.0 = ...00000000 10.2.1.0 = ...00000001
- 10.2.2.0 = ...00000010 10.2.3.0 = ...00000011
- Common prefix = 10.2.0.0 with /22 (last 2 bits vary across all four)
- Rule: 4 networks of /24 → move prefix left by 2 bits → /22

EXAM SHORTCUT — 4 networks = 2 bits of variation: /24 - 2 = /22. 8 networks = /21. 16 = /20.

Practice Question 2.6

D2 — Networking

QUESTION 2.6

A network engineer wants to advertise four contiguous /24 networks (10.2.0.0/24, 10.2.1.0/24, 10.2.2.0/24, 10.2.3.0/24) as a single route to reduce the size of the upstream routing table. What is the primary benefit of this approach and what is the correct summary route?

A. It compresses the data payload of packets passing through the network, reducing bandwidth usage.

B. It combines the four /24 routes into 10.2.0.0/22, reducing the number of routing table entries and speeding up lookups.

C. It uses UDP port 53 to resolve the four subnet names into a single DNS zone entry.

D. It dynamically reassigns IP addresses across the four subnets to eliminate address waste.

Answer — Question 2.6

D2 — Networking

ANSWER B

B. 10.2.0.0/22 — four /24s collapsed into one entry, smaller routing table.

Why this is correct

Route summarization combines contiguous networks into a single advertisement. The /22 prefix covers exactly 10.2.0.0 through 10.2.3.255 — the same address space as the four individual /24s. The result is one routing table entry instead of four: less memory, faster lookups, and reduced routing protocol traffic.

Why the other options are wrong

- A — Route summarization operates on routing table entries, not on packet payloads. It has no effect on the size or content of data packets in transit.
- C — UDP port 53 is DNS. Route summarization is a routing protocol function that has nothing to do with DNS name resolution.
- D — Summarization does not reassign or modify IP addresses. The addresses in the summarized range remain exactly as they were.